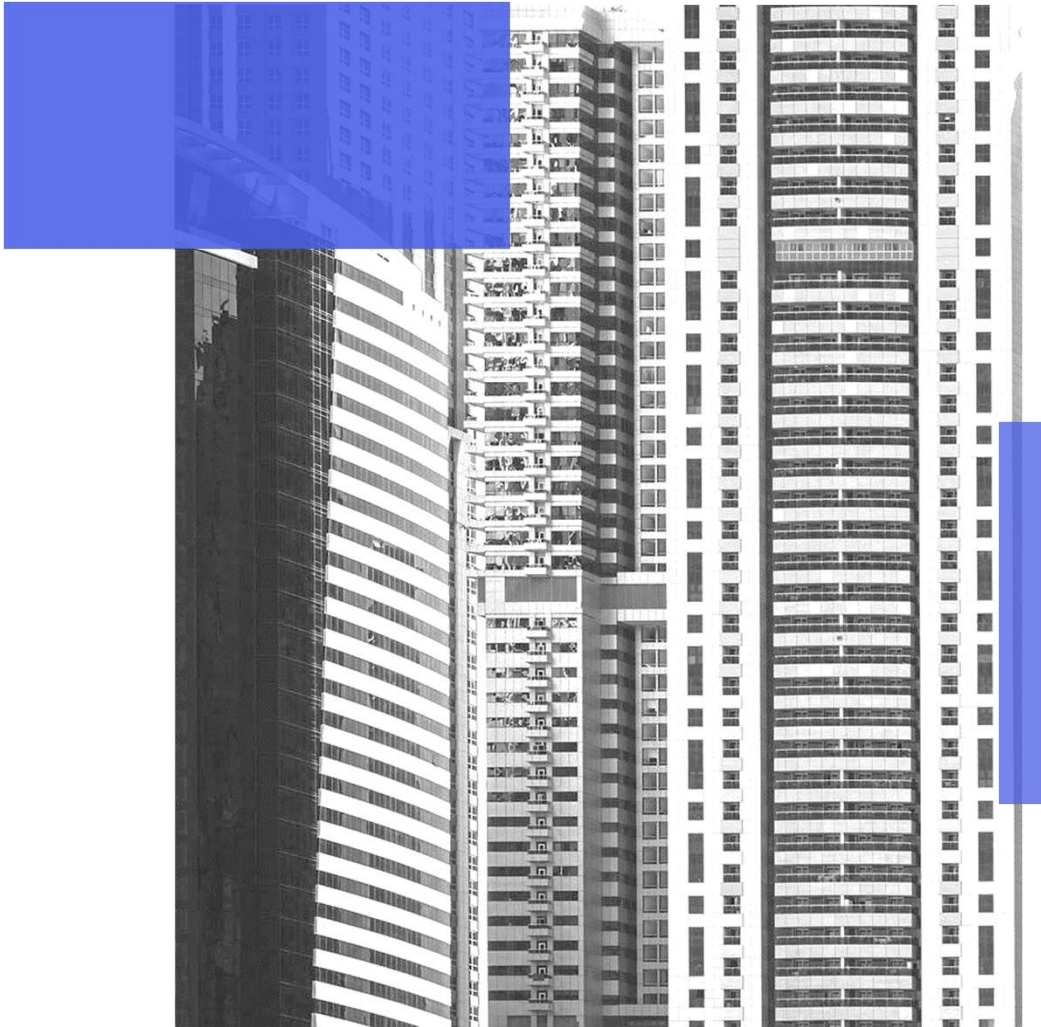


ANALYSIS AND PREDICTION

CUSTOMER CHURN





AGENDA

DATA COLLECTION

EXPLORATORY DATA ANALYSIS

MODEL DEVELOPMENT

MODEL DEPLOYMENT

PRESENTING

INTRO



Customer Churn Risk The project aims to harness data to craft a predictive model, illuminating the risk of customer churn and empowering businesses to nurture enduring customer relationships. Join us as we explore the journey of transforming raw data into actionable strategies!



DATA COLLECTION

MOHAMMAD MOSTAFA

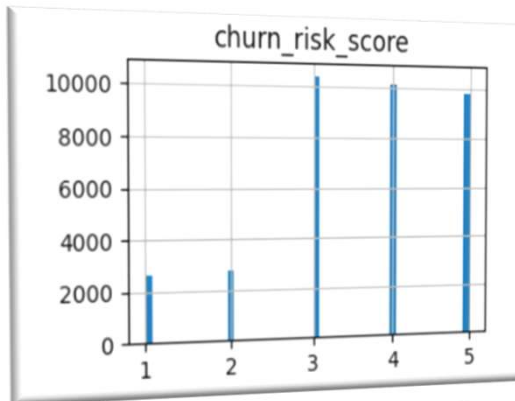
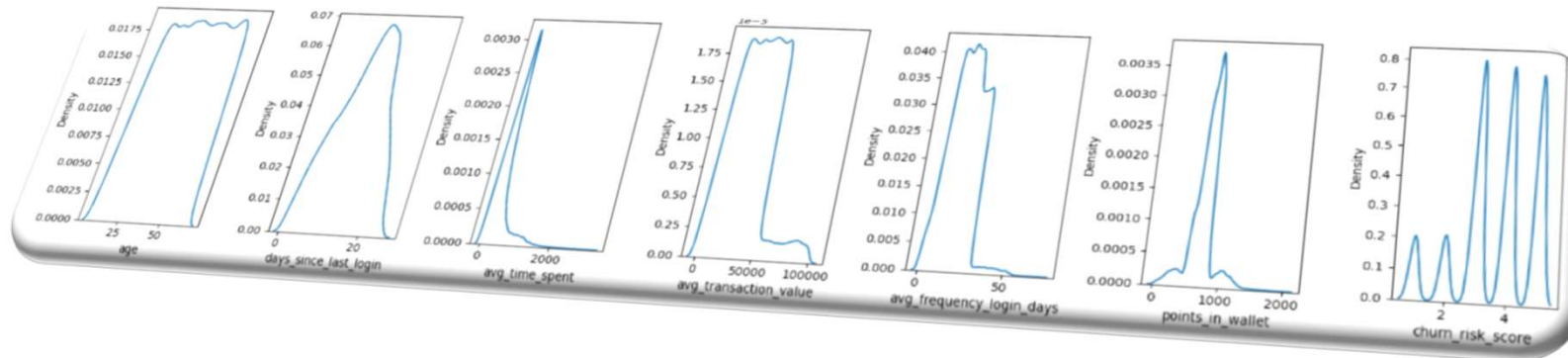
DATA

The **Customer Churn Risk Rate** dataset contains structured tabular data, including, information about customer demographics, account activity, transactional behavior, and other customer-related features that can be leveraged for churn prediction. It was originally provided as part of a HackerEarth Machine Learning Challenge aimed at predicting customer churn risk.

DATA SNIPPETS

- 36,992 rows, 25 columns
- Demographics, transactions, engagement features
- Churn risk score: -1 to 5
- Missing Values
- Imbalanced Data
- Negative values in columns
- Outliers kept for patterns

DATA SNIPPETS

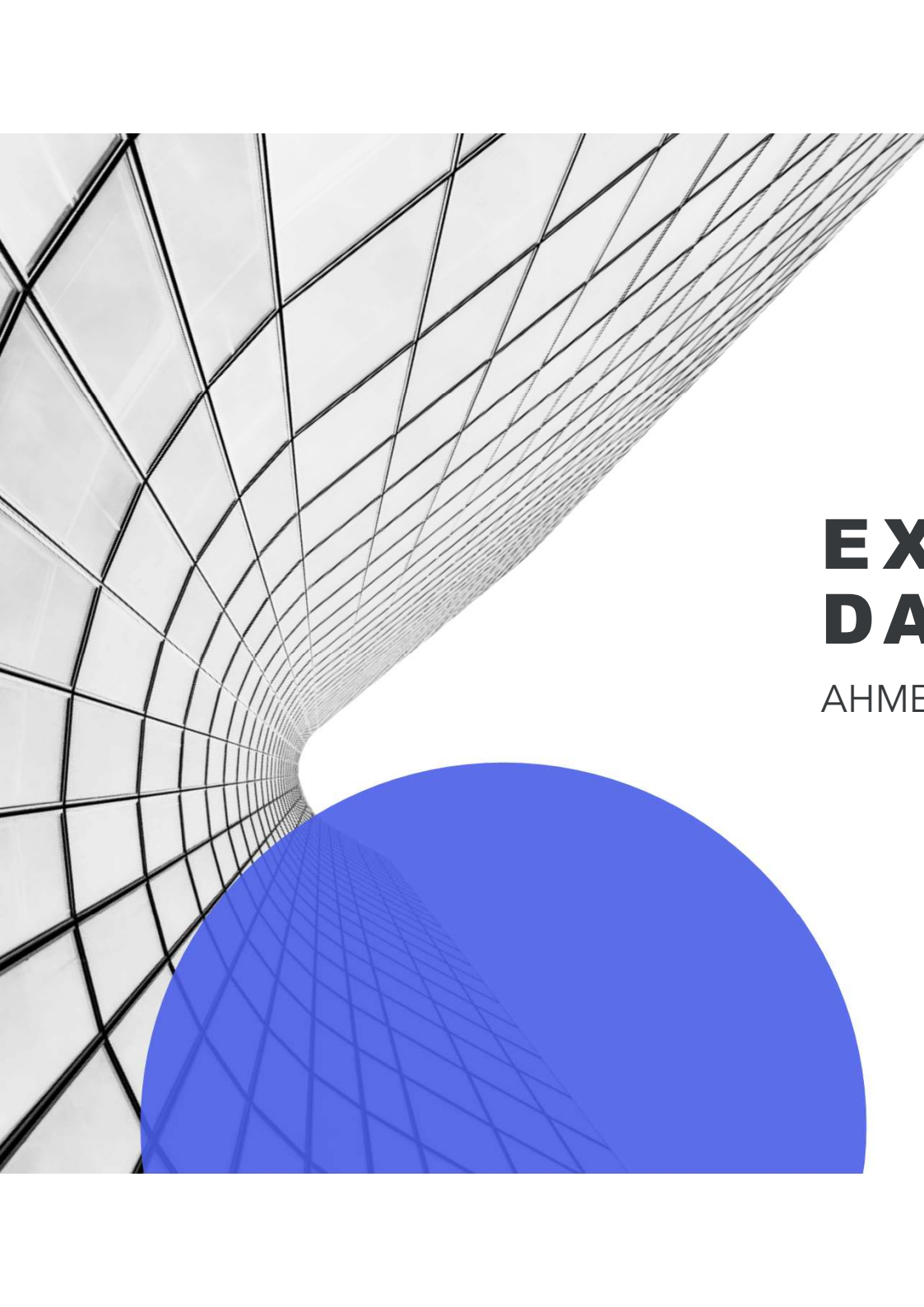


```
cleaning_pipeline = Pipeline([
    ('dataCleaner', DataCleaner(cols_to_drop, nonnegative_cols)),
    ('imputer', NaNImputerWrapper(train_sample_size=train_split.shape[0]))
])

X_train_cleaned_imputed = cleaning_pipeline.fit_transform(X_train)
X_train_split_cleaned_imputed = cleaning_pipeline.fit_transform(X_train_split)
```

CLEANING & PIPELINES

- Initial Exploration & Cleaning
- Delivering High Quality Data
- Designing an End-to-End Pipeline
- Well-Structured Notebook
- Documenting The Process



EXPLORATORY DATA ANALYSIS

AHMED SHERIF

QUESTIONS

1. Age Impact on Churn Risk?
2. Behavioral Metrics Reveal Insights?
3. Membership Levels Predict Churn?
4. Feedback Correlation with Churn?
5. Categorical Factors Matter?
6. Login Patterns Indicate Churn?

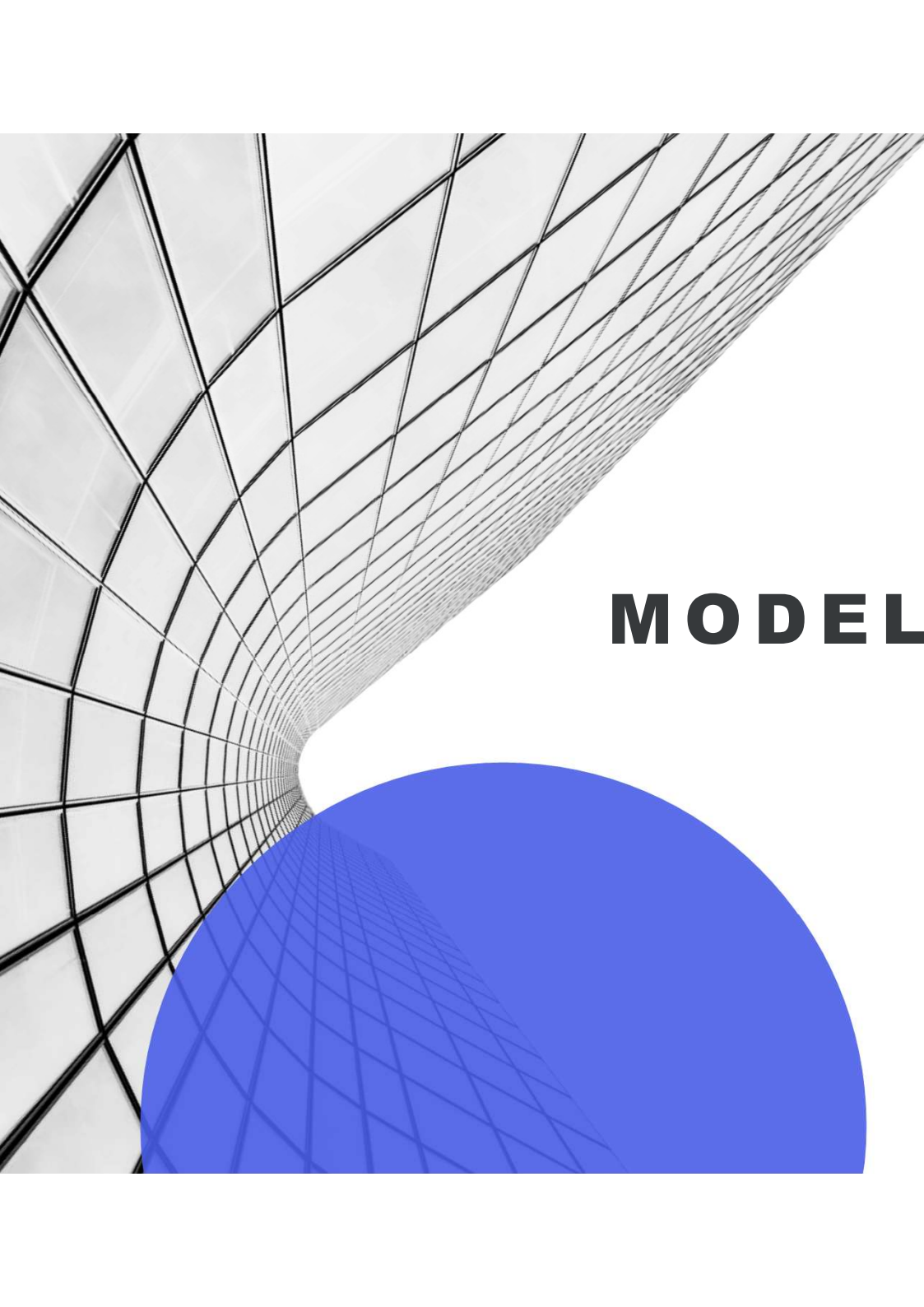
ACTIONS

1. Advanced Feature Creation
2. Categorical Feature Processing
3. Time-Based Feature Extraction
4. Feature Importance Analysis
5. Preprocessing Techniques

Wrapping All In One Pipeline

1. Create a more cohesive analytical framework
2. Ensure consistent data processing across all predictors
3. Allow for better interaction tracking between variables
4. Streamline implementation for the team
5. Make model maintenance and updates more manageable





MODEL DEVELOPMENT

AHMED M. FAYAD

INITIAL ACTIONS

1. Data Check
2. Imbalanced Classes Handled
3. Dimensionality Reduction & Feature Selection
4. Functional Programming
5. Wrapping All Pipelines In One
6. Data Pre-Processing

MODELS

1. Logistic Regression
2. Decision Tree
3. Random Forest
4. Gradient Boosting
5. XGB Boost

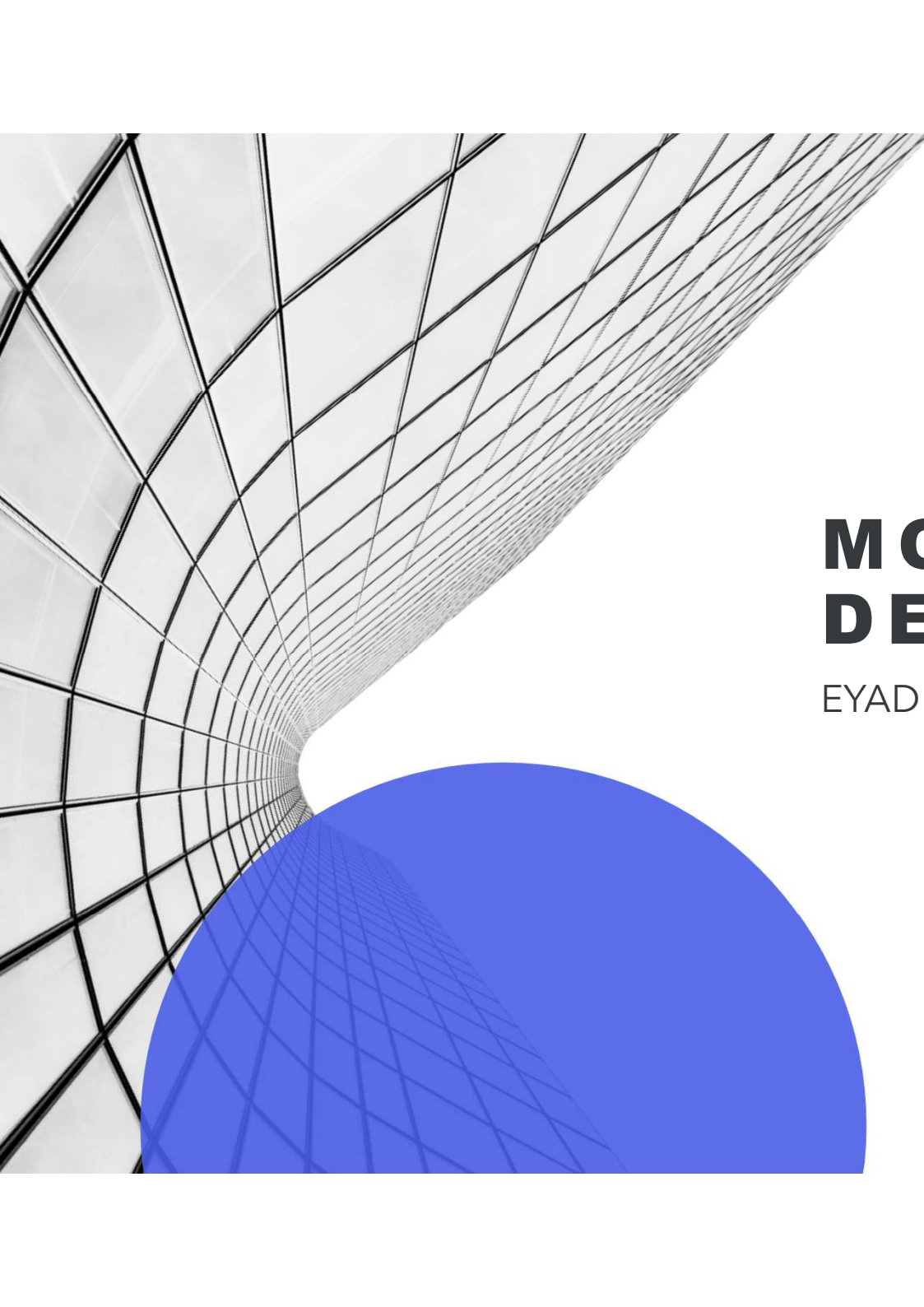
RESULTS

LR	DT	RF	GB	XGB
Accuracy: 0.6864	Accuracy: 0.6156	Accuracy: 0.6791	Accuracy: 0.6862	Accuracy: 0.6893
Precision: 0.6847	Precision: 0.6132	Precision: 0.6702	Precision: 0.6802	Precision: 0.6829
Recall: 0.6864	Recall: 0.6156	Recall: 0.6791	Recall: 0.6862	Recall: 0.6893
F1 Score: 0.6673	F1 Score: 0.6143	F1 Score: 0.6496	F1 Score: 0.6665	F1 Score: 0.6673
ROC-AUC (OVR): 0.9030	ROC-AUC (OVR): 0.7426	ROC-AUC (OVR): 0.8954	ROC-AUC (OVR): 0.9023	ROC-AUC (OVR): 0.9056

Model Tuning

1. Cross Validation Tests in Training
2. Random Search
3. Wrapping Model And Pipelines In One Object
4. Coding a Report For Results





MODEL DEPLOYMENT

EYAD AMR

DESIGNS

MANUAL PREDICTION

Entering Values Manually

BATCH PREDICTION

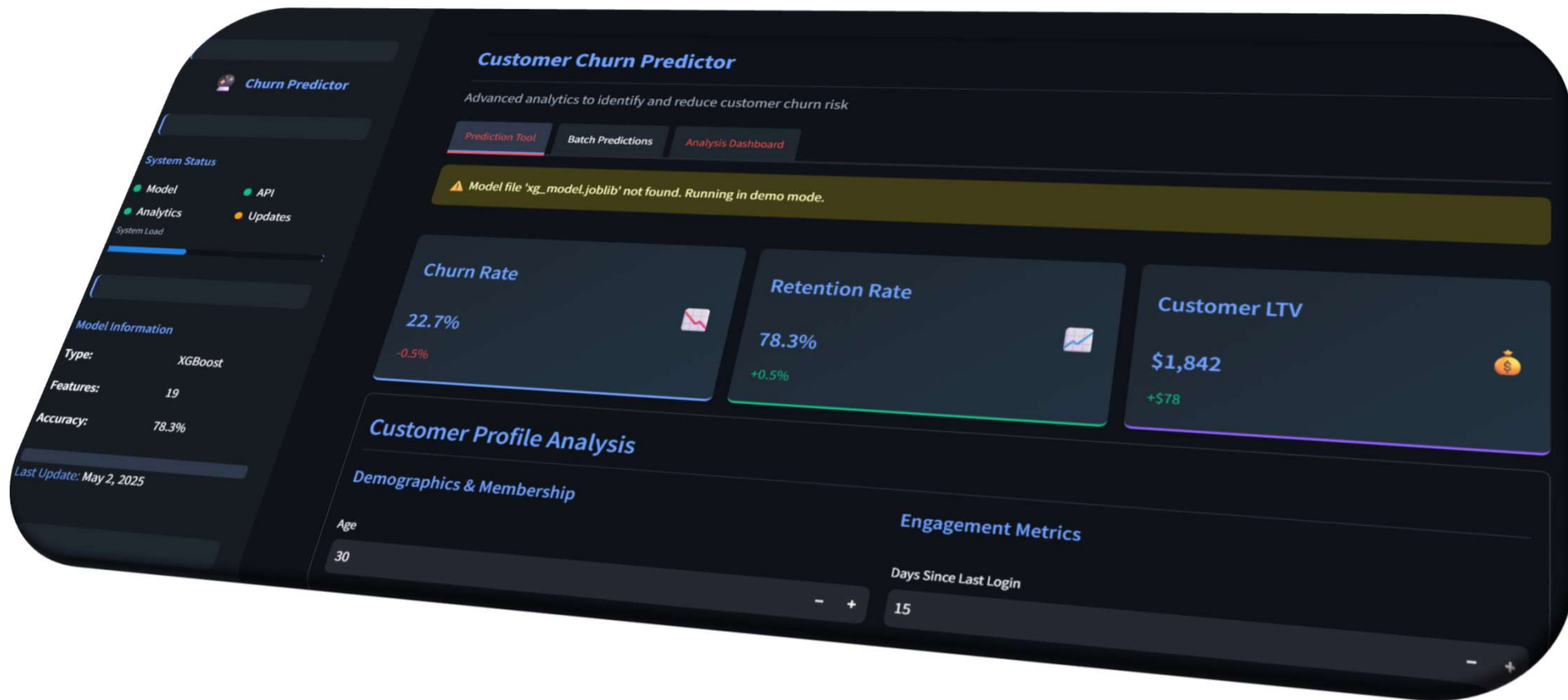
Uploading a Whole CSV File

ANALYTICS TAB

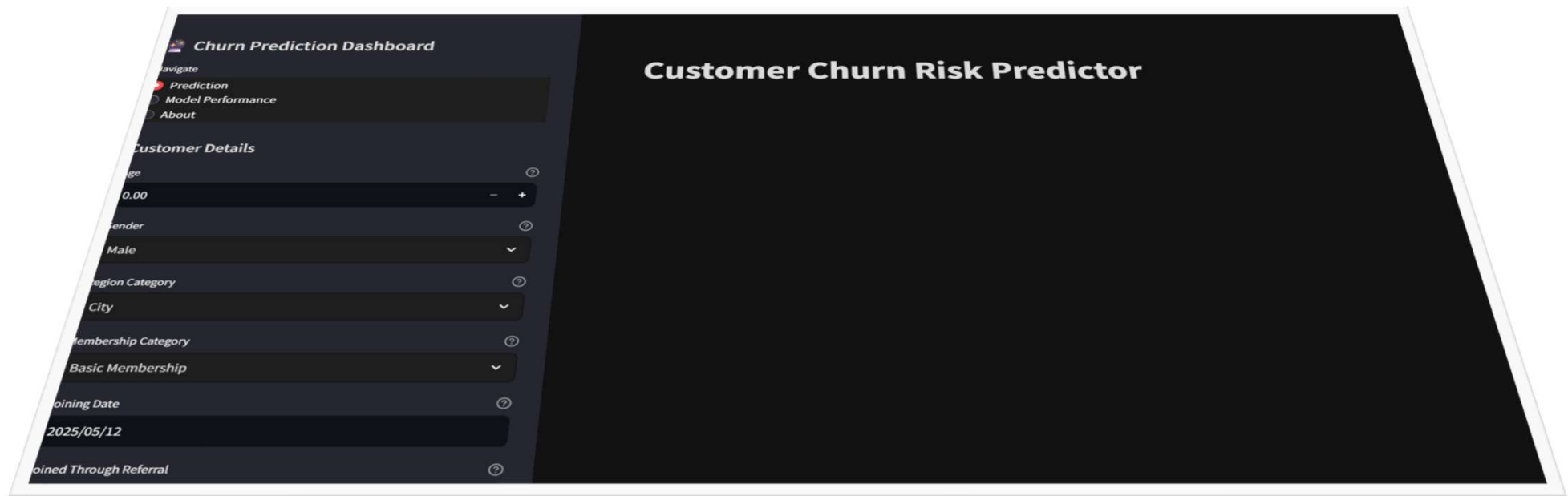
Driven Analytics From Data

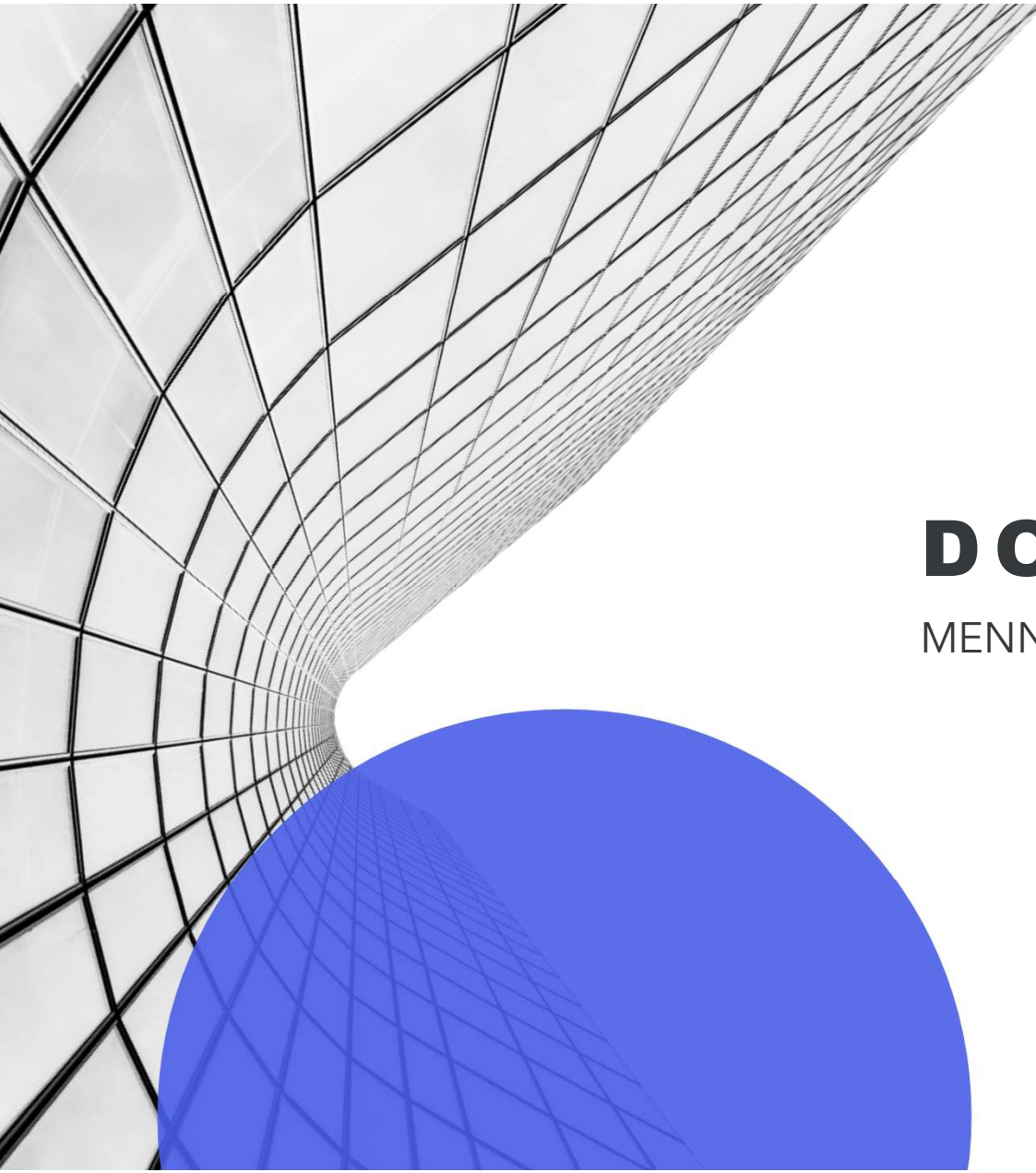


OVERVIEW



OVERVIEW





DOCUMENTATION

MENNA ELZAYAT



PLANNING THE DOCUMENTATION

- Built a centralized system to store all documentation in one place.
- Used templates and standards for consistency across code and reports.
- Applied version control (Git) to track changes in documentation just like code.



DOCUMENTING THE TECHNICAL WORK

- Created a data dictionary that explains every variable and its meaning
- Logged all data cleaning steps with clear reasons for each change.
- Mapped out the data pipeline using diagrams and clear descriptions.

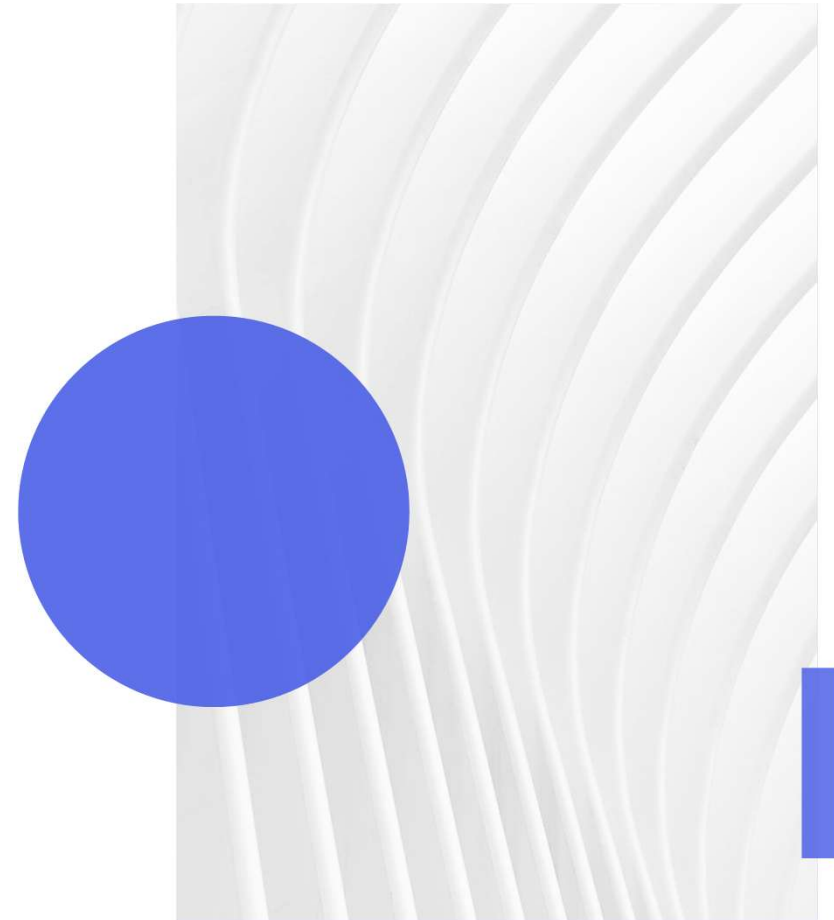


SHARING THE KNOWLEDGE

- Created training guides for both users and admins.
- Built a searchable FAQ to help with common questions.

SUMMARY

Customer Churn Risk Rate dataset, a robust and structured collection from a HackerEarth Machine Learning Challenge, brimming with customer demographics, account activities, transactional patterns, and key behavioral insights. By leveraging this rich data, our project has developed a predictive model to illuminate customer churn risk, offering businesses a powerful tool to strengthen retention strategies. This journey from raw data to actionable insights underscores the potential to foster lasting customer connections and drive sustainable growth. Thank you for joining us in this exploration!





THANK YOU